

REFRESHER COACHING 3: ELECS

February 23, 2026 | Flashing

1. The enhancement-type and the depletion-type FETs are subclasses of:
 - A. Junction FET
 - B. Metal-oxide semiconductor FETs**
 - C. BJTs
 - D. Bipolar FETs

2. With the positive probe on an NPN base, an ohmmeter reading between the other transistor terminals should be:
 - A. Low Resistance**
 - B. X
 - C. X
 - D. X

3. The main reason cascode amplifiers provide wider bandwidth than cascade amplifiers is:
 - A. X
 - B. Suppression of Miller capacitance**
 - C. X
 - D. x

4. If a circuit has an admittance of $Y = 0.5 - j0.9$, the circuit is:
 - A. X
 - B. Inductive**
 - C. X
 - D. X

5. What happens if the drain-source voltage of a MOSFET is too low in an amplifier circuit? The MOSFET:
 - A. Operates in the cutoff region
 - B. Remain in the saturation region and delivers maximum gain
 - C. Enters X
 - D. X

6. The primary difference between JFETs and depletion-type MOSFETs is: (*ChatGPT Answer*)

- A. JFETs can have positive values of V_{GS} and levels and drain current that exceeds I_{DSS} .
- B. Depletion-type MOSFETs can have positive values of V_{GS} and levels of I_D that exceed I_{DSS} .**
- C. Depletion-type MOSFETs can have only positive of V_{GS}
- D. None of the above.

7. The loaded voltage gain is _____ the no-load level.

- A. Always less than**
- B. Always greater than
- C. Usually equal to
- D. None of the choices

8. The lower cutoff frequency of a BJT amplifier can be reduced by:

- A. Decreasing capacitor values
- B. Increasing capacitor values**
- C. Reducing transistor β
- D. Increasing load resistance

$$f_c = \frac{1}{2\pi RC}$$

9. The output resistance of an ideal current source is:

- A. Zero
- B. Infinite**
- C. Very low
- D. Equal to load resistance

Ideal voltage source → zero output resistance
Ideal current source → infinite output resistance

10. In a BJT current mirror, which factor primarily affects accuracy of mirrored current?

- A. Base-emitter voltage mismatch between transistors**
- B. Collector resistor value and supply voltage fluctuations
- C. Base-emitter diode capacitance
- D. None of the above

11. The resonant frequency of parallel circuit is the same with series circuit using the same components when:

- A. The Q is very low
- B. The Q is very high**
- C. There is no resistance
- D. None of the choices

12. Replacing a standard transistor with a Darlington pair in an emitter follower will cause the value of the input impedance for the circuit to ____.

- A. Increase**
- B. Decrease
- C. Remain the same
- D. Equal to the value of Beta

13. For a given input signal, the minimum output signal voltage of an active positive clamper is equal to the ____.

- A. Value of the output
- B. Peak value of the input
- C. Voltage connected to the + input of the op-amp**
- D. Voltage connected to the - input of the op-amp

14. The common base amplifier is characterized as having a relatively ____ input impedance and relative ____ output impedance

- A. Low ; high**
- B. Low ; low
- C. High ; low
- D. High ; high

15. MOSFETs typically have an input impedance value that is:

- A. Higher than the JFET**
- B. Lower than the JFET
- C. Equal to the JFET
- D. Randomly defined relative to the JFET

16. To make a basic instrumentation amplifier, it takes ____.

- A. One op-amp with a certain feedback arrangement
- B. Two op-amps and seven resistor
- C. Three op-amps and seven capacitors
- D. Three op-amps and seven resistors**

17. In CMOS, noise margin improves when:

- A. X
- B. Supply Voltage increases**
- C. X
- D. X

18. Generally it is a good design practice for linear amplifiers to choose the operating point that is approximately ____.

- A. X
- B. X
- C. In the center of the active region.**
- D. X

19. What is the method of damping for a moving-coil instrument?

- A. X
- B. X
- C. Eddy current**
- D. X

20. Which technique reduces dynamic power in CMOS circuits?

- A. Increasing supply voltage
- B. Reducing load capacitance or clock frequency**
- C. Using longer channel transistors
- D. Increasing switching activity

21. CMOS logic gates are preferred over NMOS-only logic gates because:

- A. X
- B. They dissipate less power**
- C. X
- D. X

22. At very high temperatures the extrinsic semiconductors become intrinsic because ____.

- A. X
- B. Band to band transition dominants over impurity ionization**
- C. X
- D. X

23. When the collector resistor in a CE amplifier is increased in value, the voltage gain ____.

- A. Increases**
- B. Decreases
- C. Is not affected
- D. Become erratic

24. For the differential amplifier circuit if an input signal is applied to one of the inputs and the other connected to ground such operation is referred to as _____.

- A. **Single-ended**
- B. Double-ended
- C. Common-mode
- D. None of the above

25. The output logic levels of CMOS are:

- A. X
- B. **Nearly equal to the supply rails**
- C. X
- D. X

26. Coupling capacitor are chosen to ensure that the values of X_c are _____ at the amplifier's operating frequency.

- A. **very small**
- B. very large
- C. small
- D. Large

27. The current in an RL differentiator reaches its maximum possible value when:

- A. $5_T = t_w$
- B. **$5_T > t_w$**
- C. $5_T > t_w$
- D. $5_T = 0.5 t_w$

28. If the capacitor from emitter to ground in a common emitter amplifier is removed, the voltage gain _____.

- A. Increases
- B. **Decreases**
- C. Becomes erratic
- D. Remains the same

29. The value of drain current is always _____ the value of the short circuit drain current (I_{DSS}) for a given JFET.

- A. less than
- B. equal to
- C. **less than or equal to**
- D. greater than

30. CMOS differential amplifiers are used when _____ is desired.

- A. The highest gain
- B. Low power dissipation**
- C. Lower input impedance
- D. All of the above

31. A varactor diode is used in a frequency tuner. If the control voltage is increased from 1V to 5V, what happens to the oscillator frequency?

- A. Frequency decreases
- B. Frequency remains constant
- C. Frequency increases**
- D. Oscillator stops

32. The main reason cascode amplifiers provide wider bandwidth than cascade amplifiers is:

- A. Higher power gain
- B. Suppression of Miller capacitance**
- C. Use of feedback networks
- D. Lower bias current

33. What is the maximum output voltage swing of an op-amp?

- A. +V to -V (supply voltage)
- B. $+V_{SAT}$ to $-V_{SAT}$**
- C. + 1/2V to -1/2V
- D. Depends on the input signal

34. The positive and negative portions of a differentiator's output voltage are maximum when:

- A. $5_T < t_w$**
- B. $5_T > t_w$
- C. $5_T = t_w$
- D. $5_T > 0$

35. Cascode amplifiers are especially useful in:

- A. Low-frequency audio systems
- B. High-frequency and RF circuits**
- C. Power amplifiers only
- D. Oscillators only

36. If a bypass capacitor opens, the value of r_e will:

- A. Increase**
- B. Decrease
- C. Remain the same

D. Go to zero

37. The overall gain of a cascade amplifier is:

- A. The average of individual gains
- B. The product of individual gains**
- C. The difference of individual gains
- D. Always equal to one stage gain

38. When BJTs and FETs are used together in differential amplifiers they are called _____.

- A. BiFET circuits
- B. BiCMOS circuits**
- C. CMOS
- D. None of the choices

39. Which circuit conditions would indicate that a bypass capacitor is open?

- A. The presence of a DC voltage at the BJT's emitter terminal
- B. The voltage gain increases significantly
- C. The loss of the AC signal at the base terminal of the BJT**
- D. None of the choices

40. A BJT is in active mode. If the collector-emitter voltage is very small, approaching the saturation voltage, what happens to the current gain?

- A. Current gain increases
- B. Current gain decreases**
- C. Current gain remains constant
- D. Current stops flowing

41. The self-bias configuration develops the controlling gate-to-source voltage across a resistor introduced:

- A. In the drain leg
- B. In the gate leg
- C. In the source leg**
- D. None of the choices

42. The r'_e transistor model replaces the base-emitter junction by:

- A. a constant voltage
- B. an open circuit
- C. The AC resistance of the forward-biased diode at the operating point**
- D. a diode

43. Which is a disadvantage of a cascaded amplifier configuration?

- A. Increased overall gain
- B. Improved selectivity
- C. Increased complexity and noise**
- D. Increased gain-bandwidth product

44. What limits the bandwidth of a common-source FET amplifier?

- A. High input resistance of the gate
- B. The gate-source capacitance**
- C. The DC biasing resistors only
- D. The drain-source resistance

45. For levels of gate-to-source voltage greater than the threshold voltage, the drain current is directly related to the:

- A. square of the difference between the gate-to-source voltage and the threshold voltage**
- B. gate-drain voltage
- C. square of the gate current
- D. None of the choices

Mga naalala ko lang - bossing

46. In an Edison Battery, what is the negative plate of a NiFe battery is made of:

- A. Nickel
- B. Iron Oxide**
- C. Zinc
- D. Carbon

47. In Batteries, Electrolyte hydrolysis generally _____ the amount of usable electrolyte because some of it decomposes into gases or other byproducts.

- A. Increases
- B. Decreases**
- C. Stabilizes
- D. Has no effect on

48. h_{12} represents:

- A. Input impedance with output short-circuited
- B. Forward current gain with output short-circuited

C. Reverse voltage gain with input open-circuited

D. Output admittance with input open-circuited

49. In hybrid two-port models, h_f represents:

A. Input impedance with output short-circuited

B. Reverse voltage gain with input open-circuited

C. Forward current gain with output short-circuited

D. Output admittance with input open-circuited

50. In the h-parameter transistor model, h_{12} represents:

A. h_{ie}

B. h_{fe}

C. h_{re}

D. H_{ro}

60. A _____ is a combination of circuit elements, properly chosen, that best approximate the actual behavior of a semiconductor device under specific operating conditions.

A. circuit

B. schematic

C. model

D. monolithic IC

61. An enhancement-mode FET operates in:

A. Depletion mode only

B. Enhancement mode only

C. Both depletion and enhancement mode

D. Cutoff mode only

61. An depletion-mode FET operates in:

A. Depletion mode only

B. Enhancement mode only

C. Both depletion and enhancement mode

D. Cutoff mode only

62. The r_e is:

A. A fixed DC resistance

B. Voltage regulator

C. The dynamic (small-signal) AC resistance of a diode or emitter junction

D. Output impedance

63. Miller effect in a voltage amplifier primarily affects:

- A. Low-frequency response
- B. High-frequency response**
- C. DC bias stability
- D. Power dissipation

64. If the effective capacitance at the input of an amplifier becomes larger, the high-frequency response will:

- A. Increase
- B. Decrease**
- C. Remain unchanged
- D. Oscillate

65. The saturation voltage of a feedback (Sziklai) amplifier is approximately:

- A. 0.6 V**
- B. 0.9 V
- C. 1.2 V
- D. 2.0 V

66. The saturation voltage of a darlington amplifier is approximately:

- A. 0.6 V
- B. 0.9 V**
- C. 1.2 V
- D. 2.0 V

67. The base–emitter voltage of a Darlington pair is approximately:

- A. 1.2 V**
- B. 0.6 V
- C. 1.8 V
- D. 0.9 V

68. In a series RLC circuit at resonance, the circuit behaves as:

- A. Resistive**
- B. Capacitive
- C. Inductive
- D. Reactive

69. In a series RLC circuit, when the operating frequency is higher than the resonance frequency, the circuit behaves as:

- A. Resistive
- B. Capacitive
- C. Inductive**
- D. Reactive

70. In a parallel RLC circuit, when the operating frequency is lower than the resonance frequency, the current:

- A. Leads the voltage by 90°
- B. Leads the voltage by 45°
- C. Lags the voltage by 90°**
- D. Lags the voltage by 45°

71. Given the admittance $Y = 1 + 0.7j$, the circuit is:

- A. Resistive
- B. Inductive
- C. Capacitive**
- D. Reactive

72. Given the admittance $Y = 1 - 0.4j$, the circuit is:

- A. Resistive
- B. Inductive**
- C. Capacitive
- D. Reactive

73. In a current mirror, the source FET should operate in:

- A. Saturation**
- B. Active
- C. Cutoff
- D. Ohmic

74. In a BJT current mirror, the transistors should operate in:

- A. Saturation
- B. Active (linear)**
- C. Cutoff
- D. Ohmic

75. Comparing the common collector and common emitter configurations, the input impedance of _____ is much lower and the output impedance of _____ is much higher.

- A. Collector, emitter
- B. Collector, collector**
- C. Emitter, collector
- D. Emitter, emitter

76. The α of a BJT is:

- A. 10 to 10,000
- B. 1 to 100
- C. Typically about 0.99 but always less than 1**
- D. Typically about 10 but always less than 10000

77. An amplifier configuration where the input signal is led to the emitter terminal and the output from the collector terminal is called

- A. Common base**
- B. Common emitter
- C. Clipper
- D. Common collector

78. The iterative impedance of a two-port network terminated with that same impedance is also called:

- A. Characteristic impedance**
- B. Image impedance
- C. Input impedance
- D. Dynamic impedance

79. The voltage at the cutoff frequency of an amplifier is also called:

- A. -6 dB
- B. -3 dB**
- C. Unity gain point
- D. Half-voltage point

80. The voltage at the cutoff frequency of an amplifier is:

- A. 50% of the gain
- B. 70.7% of the gain**
- C. Equal to the maximum gain
- D. Zero

81. In a field-effect transistor, the transconductance g_m under practical operating conditions is _____ the g_{m0} (maximum transconductance).

- A. Less than

- B. Greater than
- C. Less than or equal to**
- D. Independent of

82. In CMOS logic, an inverter is implemented using:

- A. Two NMOS transistors
- B. Two PMOS transistors
- C. PMOS and NMOS transistors**
- D. Bipolar junction transistors

83. Which logic family is known for fast switching speed?

- A. CMOS**
- B. PMOS
- C. NMOS
- D. VMOS

84. An asymmetrical network whose electrical characteristics are not identical when measured from both ports is classified as:

- A. Balanced
- B. Unbalanced**
- C. Iterative
- D. Symmetrical

85. In a unijunction transistor (UJT), when the channel resistance decreases, the base voltage _____ and the emitter voltage _____.

- A. Decrease, decrease**
- B. Increase, increase
- C. Decrease, increase
- D. Increase, decrease

86. Rise time is defined as the time required for a signal to change from:

- A. 0% to 50%
- B. 10% to 90%**
- C. 1% to 100%
- D. 95% to 99%

87. In a common-base amplifier, the configuration provides:

- A. Voltage gain and power gain but no current gain**
- B. Voltage gain, power gain, and current gain
- C. Current gain and power gain but no voltage gain

D. Current gain only

88. In a common-collector amplifier, the configuration provides:

- A. Voltage gain and power gain but no current gain
- B. Voltage gain, power gain, and current gain
- C. Current gain and power gain but no voltage gain**
- D. Power gain only

89. In a common-emitter amplifier, the configuration provides:

- A. Voltage gain and power gain but no current gain
- B. Voltage gain, power gain, and current gain**
- C. Current gain and power gain but no voltage gain
- D. Current gain only

90. It satisfies the conditions of unity loop gain and zero phase shift and without requiring an external input signal

- A. Rectifier
- B. Amplifier
- C. Oscillator**
- D. Voltage regulator

91. A bridge rectifier:

- A. Uses both half cycles of the input AC signal**
- B. Uses only the negative half cycle
- C. Uses only the positive half cycle
- D. Does not require diodes

92. In an ideal filter response, signals within the passband experience zero attenuation, while signals within the attenuation band is _____

- A. Low
- B. Infinite**
- C. High
- D. Zero

93. To prevent damage to sensitive electronic components due to static charges, a common protection method is the use of:

- A. Series resistors
- B. Back-to-back Zener diodes**
- C. Coupling capacitors
- D. RF chokes

94. In a linear power supply, a Zener diode is primarily used to provide:

- A. For filtering ripples
- B. Rectification
- C. Stable output voltage**
- D. Power factor correction

95. Used to provide isolation between the input and output is the:

- A. Rectifier
- B. Transformer**
- C. Filter capacitor
- D. Zener diode

96. Cobalt is classified as:

- A. Diamagnetic
- B. Ferromagnetic**
- C. Paramagnetic
- D. Non-magnetic

97. In a current mirror, the transistors used are typically:

- A. Different
- B. Identical**
- C. Randomly selected
- D. Unmatched

98. A current mirror is most affected by:

- A. Supply voltage variations
- B. Mismatch between the base-emitter voltages**
- C. Load resistance
- D. Transistor parameter variations

99. A CMOS integrated circuit

- A. Can only work at low frequencies
- B. Requires considerable power to function
- C. Is susceptible to damage by static**
- D. Needs very high voltage

100. The h-parameter that is the equivalent of the β of a common-emitter circuit is _____.

- A. H_{ie}
- B. h_{re}

- C. h_{fe}
- D. h_{ro}

101. The feedback pair and the Darlington pair are very similar to each other. One difference between them is _____.

- A. there is no difference between them
- B. there is no similarity between them
- C. **the feedback pair uses one npn and one pnp transistor**
- D. the Darlington pair uses one npn and one pnp transistor

102. The feedback pair is similar to the Darlington circuit but it is _____.

- A. more used
- B. **more complex**
- C. simpler to analyze
- D. None of the above

103. Applying an input signal to one of the inputs of differential amplifier and connecting the other input to ground, you create a _____ configuration.

- A. **single-ended**
- B. common-mode
- C. double-ended
- D. None of the above

104. Applying equal input signals to both inputs of a differential amplifier results in _____ configuration.

- A. single-ended
- B. common-mode
- C. **double-ended**
- D. None of the above

105. The common-base amplifier is characterized as having a relatively _____ input impedance and relatively _____ output impedance.

- A. high; high
- B. high; low
- C. low; low
- D. **low; high**

106. If the resistor in the emitter leg is not bypassed by a capacitor then the input impedance of the small signal amplifier will _____.

- A. decrease

- B. increase**
- C. stay the same
- D. increase in some cases and decrease in other cases

107. The input impedance of a BJT is _____.

- A. inductive
- B. resistive and capacitive
- C. capacitive
- D. Resistive**

108. Depending on the configuration of the amplifier, the magnitude of the no-load voltage gain for a single BJT transistor amplifier typically ranges from _____.

- A. 10 to about 10,000
- B. just a little less than 1 to a few hundred**
- C. a hundred to about a million
- D. None of the above

109. The h-parameter model uses _____ parameters to describe the equivalent circuit of the BJT transistor.

- A. two
- B. three
- C. four**
- D. Five

110. The h-parameter model and the re parameter models are almost identical if the parameter _____ in the h-parameter model is ignored.

- A. h_f
- B. h_i
- C. h_r**
- D. h_o

111. The approximation that allows superposition to be used to isolate the ac analysis and the dc analysis of small-signal amplifiers is that the circuit response is _____.

- A. Non-linear
- B. linear**
- C. dc non-linear and ac linear
- D. dc linear and ac non-linear

112. In a lithium-ion battery, what allows ion flow while preventing short-circuit between electrodes?

- A. anode
- B. cathode
- C. separator**
- D. electrolyte